

Week 1 – Practice Session

September 7–11, 2026

In every problem, $x \in \mathbb{R}$.

Equations in One Unknown. Linear Equations. The Balance Principle.

1. $2x + 35 = 107$
2. $2x + 3 + (5x - 8) = 1 - 2x$
3. $-4(6 - 5x) + 4(8x - 9) = -7$
4. $2(x - 2) - 3(2x + 1) = 3(-2x + 3) - (x - 2) + 3$
5. $x - \frac{6 - 2x}{3} = 2x - 4 - \frac{x + 3}{2}$
6. $-4 - \frac{9x - 120}{4} = 3x + 5$

Quadratic Equations.

7. $x(x - 3) = 8(x - 3)$
8. $\frac{3x^2 - 11}{4} + \frac{74 - 2x^2}{6} = 20$
9. $x^2 - 4x = 0$
10. $2x^2 = 10$

Higher-Degree Equations Reducible to Quadratic Equations.

11. $x^4 - 5x^2 + 4 = 0$
12. $(x - 1)^4 + 3(x - 1)^2 = 10$

Fractional Equations.

13. $\frac{1}{x} + \frac{2}{3} = \frac{5}{6}$
14. $\frac{x + 1}{x - 2} = 3$
15. $\frac{2x + 1}{x - 3} + 1 = -\frac{4x - 5}{x - 3}$
16. $\frac{18x + 2}{x - 4} - \frac{15x + 1}{x + 5} = 3$